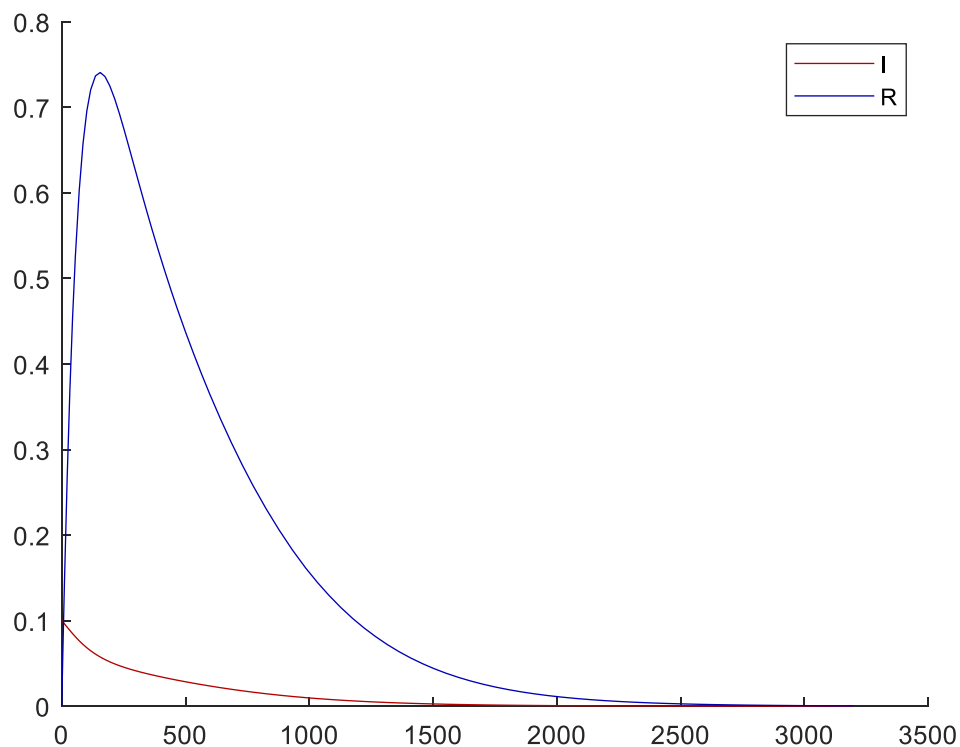


Disease free equilibrium is stable after the vaccination ( $R_{0,vac} < 1$ )

\*\*\*\*\* Abdul Parameters\*\*\*\*\*

- \*\*\*\* Equilibrium (S,I,V,R) \*\*\*\*  
18.8106    2.3442    48.8040    30.0412  
Eigenvalues  
-0.0160 + 0.0346i    -0.0160 - 0.0346i    -0.0000 + 0.0000i    -0.0027 + 0.0000i
- \*\*\*\* Equilibrium (S,I,V,R) \*\*\*\*  
11.2929    0.1675    86.3926    2.1470  
Eigenvalues  
-0.0115 + 0.0089i    -0.0115 - 0.0089i    0.0013 + 0.0000i    -0.0000 + 0.0000i
- \*\*\*\* Equilibrium (S,I,V,R) \*\*\*\*  
10.0000    0    90.0000    0  
Eigenvalues  
-0.0095    -0.0000    -0.0111    -0.0029

A small amount of imported Infected cases (like 0.1 percentage of the population) will not spread the disease and the system will go back to DFE.



But...

If the vaccination is taking after the endemic equilibrium is reached, the disease is no longer eliminated.

Endemic equilibrium is stable even with the vaccination ( $R_{0,vac} < 1$ )

\*\*\*\*\* Abdul Parameters\*\*\*\*\*

\*\* Equilibrium 1 \*\* (unstable)

```

S=100 I=0 V=0 R=0
Eigenvalues
-3.65e-05 -0.011148 0.35723 -0.00094974
** Equilibrium 2 ** (stable endemic)
S=28.5714 I=5.1703 V=0 R=66.2582
Eigenvalues
-0.018503+0.060332i -0.018503-0.060332i -3.65e-05+0i -0.0061214+0i
** Equilibrium 3 ** (unfeasible)
S=7.4349 I=-0.9495 V=105.6824 R=-12.1679
Eigenvalues
-0.025852+0i 0.0097266+0.0088635i 0.0097266-0.0088635i -3.65e-05+0i

```

